

12.1 Solving Problems With Lengths of Intersecting Chords

Lesson Introduction

 Benchmark Focus	MA.912.GR.6.1 Solve mathematical and real-world problems involving the length of a secant, tangent, segment, or chord in a given circle.		 Learning Target	I can solve mathematical and real-world problems involving lengths formed by the intersection of two chords.
 Benchmark Coherence	Building On: N/A		Working Toward: N/A	
 Essential Questions	- What is the relationship between the lengths of two intersecting chords? - What is the relationship between two chords if the two chords are equidistant from the center of a circle?			
 Lesson Narrative	This lesson activates prior knowledge of perpendicular bisectors, chords, and triangle similarity theorems. Students will be introduced to the properties of chords in circles through exploration. The primary learning target in this lesson focuses on two chords being equidistant from the center of the circle and intersecting chords.			
 Component Summary	12.1.1 Warm-Up (~5 min.)	Notice and wonder about bowling ball pyramid (<i>SE p. 231</i>)	12.1.3 Bring It Together (10 min.)	Chord relationships in circles (<i>SE p. 234</i>)
	12.1.2 Exploration (15-20 min.)	Using patty paper to explore perpendicular bisectors on chords (<i>SE p. 232</i>)	12.1.4 Guided Practice (10 min.)	Practice finding lengths of chords (<i>SE p. 236</i>)
	12.1.5 Wrap-Up (~5 min.)	Reflection on mastery of learning targets (<i>SE p. 237</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Chord - Diameter - Radius - Congruent - Perpendicular - Bisect		- Patty Paper (Optional) Colored Pencils or Markers	
			Additional Preparation	
			12.1.2 Exploration has students copying three circles and constructing their perpendicular bisectors. Prepare patty paper for each student.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may not recall the properties of a perpendicular bisector. - Students may add the intersecting chords together, rather than find the product of their segments. - Students may struggle acquiring new vocabulary/formulas and connecting them with known vocabulary/formulas.	- Provide colored markers or pencils to students to use in question 2 of 12.1.2 Exploration. They can highlight the different parts of the circle they are working with (chords, radii, perpendicular bisector, triangle). - Consider illustrating the three chord relationships on a separate piece of chart paper that can be hung in the classroom as a reference for the students.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share 12.1.2 Exploration provides an opportunity for students to use this routine. However, the questions are designed to spark thinking for the individual learners. Allow each student time to make a conjecture about the answer to the questions, share their reasoning with a partner, and have a variety of students share with the whole class.	MA.K12.MTR.1.1: Participate in Effortful Learning The expectation of 12.1.2 Exploration is to develop understanding through the use of manipulatives, visual models, discussion, estimation, and drawings. Continue to encourage students to use correct terminology when discussing with partners to help deepen their understanding.
 Differentiate Instruction	12.1.2 Exploration provides a variety of entry-points for learners to engage with the content. Part a provides a hands-on component along with an individual conjecture opportunity. Question 1b offers space for verbal reasoning with a partner, and question 1c positions students as mathematicians, expressing what they know from question 1a and 1b into a conjecture. Some students may prefer question 1a, and independently process information, while other students might prefer question 1b, and verbally process in order to make sense of the information. Ensure equitable time for all three parts during the component.	
Lesson Notes:		

12.1.1 Warm-Up: Notice and Wonder

Component Overview: Students reflect on and then share what they notice and wonder about a photograph of a pyramid of bowling balls. The photograph is intended to inspire student curiosity in the upcoming lesson. Ask students how the photograph may connect to exponents.



12.1.1 Warm-Up: Notice and Wonder

1. A picture of a bowling-ball pyramid is shown.



Notice and Wonder

Prompt students to reflect on what they notice and wonder about the photograph. Call on a variety of students to share what they noticed and wondered. This activity provides a low floor, as all students can participate, and a high ceiling, as students can be challenged to think of multiple things they may notice and wonder.

- a. What do you wonder?

Answers will vary. How many bowling balls are there? How was the pyramid made?

- b. What do you notice?

Answers will vary. The bowling balls are different colors. Each row has the same number of bowling balls.

- c. Write a question about the pyramid that a 5th grader could answer.

How many bowling balls are in the pyramid?

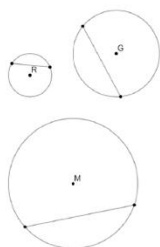
12.1.2 Exploration: Chords

Component Overview: Students will use patty paper to copy circles and then construct a perpendicular bisector on the chord inside each given circle. Students will write a conjecture and share with a partner. They will then use prior knowledge of perpendicular bisectors and similar triangle properties to explore the effects they have on the chords of a circle.



12.1.2 Exploration: Chords

1. Circles R , G , and M are shown. Each circle has a chord. Use patty paper to copy the circles. Use the patty paper copy to construct a perpendicular bisector for each chord by folding.



- a. Write a conjecture about the perpendicular bisector of a chord of a circle.

Answers will vary. The perpendicular bisector of a chord in a circle passes through the center of the circle.

- b. Share your conjecture with your partner. Listen to your partner's conjecture and write a summary. Have your partner initial that it is summarized correctly.

Summary of Partner's Conjecture	Partner's Initials
<i>Answers will vary.</i>	

- c. Review your conjecture and make any revisions, if needed.

Answers will vary.



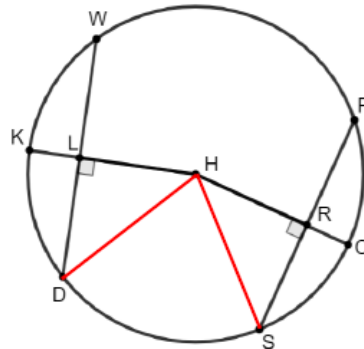
What is the difference between a chord, a radius, and a diameter? Can a chord be a radius? Why or why not? Can a diameter be a chord? Why or why not?



Teachers may need to review the construction of a perpendicular bisector. Students are not expected to have a perfect mathematical knowledge of chords and their properties at this point. This concept is being developed through exploration, so encourage informal conjectures.

12.1.2 Exploration: Chords (continued)

2. Circle H is shown, where $\overline{KH}$ and $\overline{CH}$ are radii of the circle. $\overline{WD}$ and $\overline{PS}$ are chords of the circle equidistant from the center H . Therefore, $\overline{LH} \cong \overline{RH}$.



- Using your conjecture about perpendicular bisectors, complete each congruence statement.
 $\overline{DL} \cong \overline{WL}$
 $\overline{PR} \cong \overline{SR}$
- Draw a line segment from point D to point H and a line segment from point S to point H .
See above.
- Justify why $\overline{DH}$ and $\overline{SH}$ are congruent.
They are congruent because they are both radii of the circle.
- Given that $\overline{LH} \cong \overline{RH}$ and $\overline{SH} \cong \overline{DH}$, what triangle congruence theorem justifies that $\triangle DLH$ and $\triangle SRH$ are congruent?
Hypotenuse Leg Theorem
- Since $\triangle DLH \cong \triangle SRH$, what is true about $\overline{DL}$ and $\overline{SR}$?
They are congruent and therefore have equal measures.
- What is true about chords $\overline{DW}$ and $\overline{SP}$?
They are congruent and therefore have equal measures.



Notice and Wonder

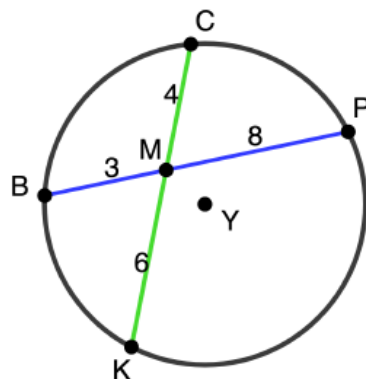
Consider providing students a few minutes to write down things they notice and wonder about the image before answering the questions. This will help students get familiar with the circle and chords prior to answering the questions.



Create a new visual aid, chart paper sized or interactive notebook sized, with a large circle, illustrating several of the vocabulary words. Because students are already familiar with some of the vocabulary words, try a word bank activity and have students label the different parts, such as the radius, diameter, center, and chord.

12.1.2 Exploration: Chords (continued)

3. Circle Y is shown, where points C , P , K , and B are on circle Y . Chords $\overline{PB}$ and $\overline{KC}$ intersect at point M .



Observations of the Blue Chord Measures	Observations of the Green Chord Measures
$3 \times 8 = 24$	$6 \times 4 = 24$
Commonalities in the Blue and Green Chords <i>Answers will vary. The products of the two pieces of the chords both equal 24.</i>	
What do you wonder about the relationship between the blue chord and the green chord? <i>Answers will vary. Will all intersecting chords of a circle cut the chords into two pieces that when multiplied will equal the same product?</i>	



For struggling students, consider providing a chart for students to look at to see the different possible operations for each chord measure.

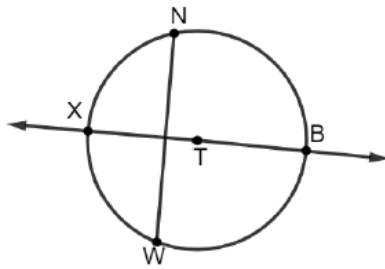
12.1.3 Bring It Together: Chord Relationships

Component Overview: Students will work individually, utilizing 12.1.2 Exploration as a guide. They will be introduced to the properties of the diameter of a circle that is perpendicular to a chord, and chords equidistant from the center. This component serves as a connection to their findings from 12.1.2 Exploration. Encourage students to use the correct vocabulary when discussing these new properties.



12.1.3 Bring It Together: Chord Relationships

- Circle T is shown with chord $\overline{NW}$ and secant $\overline{XB}$, where $\overline{XB}$ bisects $\overline{NW}$ and $\overline{XB}$ is a diameter of the circle.



What information is known about circle T , chord $\overline{NW}$, and the angles formed by $\overline{XB}$ and $\overline{NW}$?

Our previous conjecture stated that if a diameter bisects a chord, then the diameter is perpendicular to the chord so $\overline{XB} \perp \overline{NW}$.



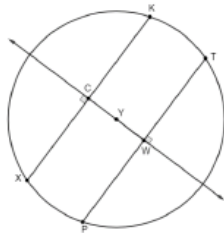
If a diameter of a circle is perpendicular to a chord, then the diameter bisects the chord and its arc.



Consider using students' conjectures from 12.1.2 Exploration to guide student understanding of the relationships between chords and perpendicular bisectors.

12.1.3 Bring It Together: Chord Relationships (continued)

2. Circle Y is shown with chords $\overline{XK}$ and $\overline{PT}$, where $\overline{CY} \cong \overline{WY}$.



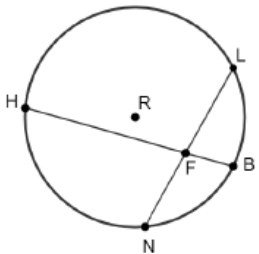
What information is known about circle Y , chords $\overline{XK}$ and $\overline{PT}$, and the angles formed by $\overline{CW}$?

Since $\overline{CY} \cong \overline{WY}$, then $CY = WY$, therefore showing the two chords are equidistant from the center of circle Y . Since distance is the perpendicular distance then the two chords are congruent.



If two chords are equidistant from the center of a circle, then the chords are congruent.

3. Circle R is shown with chords $\overline{HB}$ and $\overline{LN}$ that intersect at point F .



What information is known about circle R , chords $\overline{HB}$ and $\overline{LN}$, and the angles formed?

$$(HF)(FB) = (LF)(FN)$$

$$m\angle HFN = m\angle LFB \text{ and } m\angle HFL = m\angle NFB$$



When two chords intersect each other inside a circle, the products of their segments are equal.



Allow students to engage in meaningful dialogue related to the properties of chords in questions 1-3 to ensure they understand each relationship and the difference between the properties.



Prompt students to make sense of this information in their own way. Put students in pairs to construct an example that illustrates the meaning of the properties in each blue box.

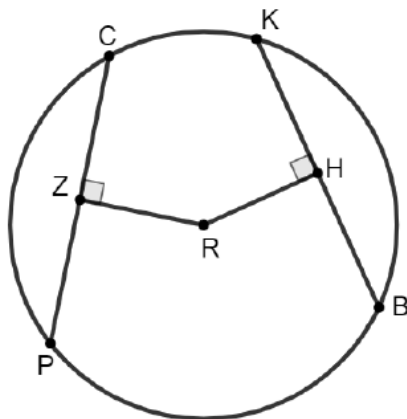
12.1.4 Guided Practice: Solving Problems Involving Intersecting Chords

Component Overview: Students will apply their conceptual understanding to solve questions involving intersecting chords. Students can complete this component independently and peer review their partners' responses.



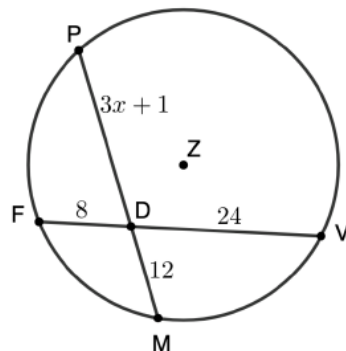
12.1.4 Guided Practice: Solving Problems Involving Intersecting Chords

- Circle R with points C , K , B , and P on the circle are shown with chords $\overline{CP}$ and $\overline{KB}$ equidistant from the center. Find the value of x , if $CP = 6x - 7$, $KB = 35$, $ZR = HR$, and $m\angle CZR = m\angle KHR = 90^\circ$.



$$6x - 7 = 35, \quad x = 7$$

- In circle Z , chords $\overline{PM}$ and $\overline{FV}$ intersect at point D . Points P , F , M , and V lie on circle Z . Given $FD = 8$, $DM = 12$, $VD = 24$, and $DP = 3x + 1$, solve for x .
 $(3x + 1)(12) = 8(24)$
 $x = 5$



Use informal observations of student responses from 12.1.2 Exploration to inform how students will engage with this component.

- Students can collaborate on question 1 in pairs, using a Turn and Talk routine to facilitate discussion. In pairs, students then choose either question 2 or question 3 to complete before comparing responses.
- The teacher facilitates a student-centered discussion on question 1 before releasing students to attempt questions 2-3 on their own.
- The teacher pulls a small group for accelerated support, while the rest of the class independently attempts questions 1-3.



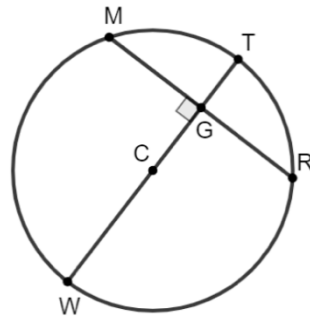
Describe the steps for solving a problem involving the lengths of intersecting chords within a circle. Create/produce an example for use in your explanation.

12.1.4 Guided Practice: Solving Problems Involving Intersecting Chords (continued)

3. Circle C is shown, where points T , R , W , and M are on the circle. The diameter, WT , is a perpendicular bisector of chord MR . If $MG = (y + 4)$ and $GR = (2y + 1)$, what is the length of the chord MR ?

$$y + 4 = 2y + 1 \quad y = 3$$

$$MR = 14$$



Will the intersecting chords theorem work if one of the chords is a diameter? Why or why not? Explain.



Consider offering students the option to color code each chord and their lengths given. This will help students identify which measures go with each chord.



12.1.5 Wrap-Up: Reflection

Answers will vary.

- | | | |
|----------------------|------------------------|-----------------------|
| I need help. | I'm getting there, but | I understand this |
| I can't get started. | I need more practice. | and I feel confident. |

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